

Engineering Package Output Structure

engineering_package/

├─ stl_parts/ (38 STL meshes)

├─ step_parts/ (38 STEP B-rep models)

├─ freecad_scripts/ (38 parametric .py scripts)

├─ ros2_package/

│ └─ urdf/ (URDF + flat URDF)

│ └─ meshes/ (38 STL for simulation)

│ └─ launch/ (display + gazebo launch)

│ └─ config/ (ros2_control + joint_names)

├─ firmware/ (6 files: servo, DC motor, IK, odometry)

├─ assembly.stl (16MB, with fastener geometry)

├─ urdf.xml (49KB, MuJoCo-ready)

├─ bom.md (bill of materials with COTS P/N)

├─ assembly_guide.md (25KB step-by-step)

├─ design_report.json

├─ cable_routing_report.md

├─ power_report.md

├─ wiring_diagram.md

└─ README.md

Total: 186 files

.stl: 77 files

.py: 42 files

.step: 38 files

.md: 7 files

.json: 5 files

.cpp: 4 files

.xml: 3 files

.h: 3 files

.yaml: 2 files

.urdf: 2 files

.ino: 1 files

.txt: 1 files

.rviz: 1 files